

STATISTICAL ANALYSIS AND PRACTICAL INSIGHTS ON CROP

An Internship Report

Submitted in partial fulfillment of the requirements

for the award of the degree of

MASTER OF SCIENCE

in

STATISTICS

By

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in

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JULY 2025

CERTIFICATE



ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the Department of Economics and Statistics (DES), Government of Tamil Nadu, for providing me the invaluable opportunity to undertake this internship. The experience of working with real-time data and being part of meaningful fieldwork has been immensely enriching. I am especially thankful to my field supervisor and the statistical officers at DES for their continuous guidance, patience, and support throughout the internship period.

I also extend my heartfelt thanks to the faculty members of the Department of Statistics, University of Madras, for their constant academic support, mentorship, and encouragement throughout the course of my study and this internship project.

Finally, I am grateful to all the respondents and field staff who contributed their time and perspectives, making this research possible.

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Introduction

This dataset provides granular agricultural statistics across Indian districts from the year 1997 onward, covering a wide variety of crops grown in different seasons. It consists of 2,753 records and includes variables related to crop type, area under cultivation, production volume, and yield rates. Each entry is categorized by state, district, crop year, and growing season. The dataset captures essential agricultural metrics: Area (in hectares) Production (in tonnes) Yield (in kg/hectare), both raw and rounded. The presence of calculated yield (Yield) and its facilitates quick comparative analyses and visual representation. With detailed crop-level data, this dataset is valuable for conducting productivity analysis, studying the effects of policy or climate on agriculture, and identifying regional trends and disparities in agricultural output. Researchers, policymakers, and data analysts can utilize this dataset. Compare productivity across crops, districts, and years. Detect yield anomalies (spikes or drops). Support agricultural planning and intervention strategies.

Objectives

1. Statistical Analysis
2. Crop-wise Statistical Comparisons
3. Trend and Growth Analysis Over Time
4. Crop-specific Performance Analysis
5. Efficiency & Productivity Metrics
6. Comparative Performance Between Crops
7. Time Series Decomposition and Smoothing
8. Abrupt Changes & External Factors

About the dataset

The dataset provides district-level agricultural statistics for Tamil Nadu, focusing on crops across multiple years. This dataset can be used to analyze changes in agricultural output, compare crop performance across years, and explore factors influencing production and yield variations.

Methodology:

1. Data Collection

The data has been sourced from official agricultural records maintained by state or national authorities, such as the Directorate of Economics and Statistics (DES), Ministry of Agriculture & Farmers Welfare, Government of India. These records are typically compiled from field-level reports submitted by local agricultural officers at the district level.

2. Variables Captured

1. The dataset includes the following key variables:
2. Year: The agricultural year (e.g., 2010, 2011, etc.)
3. State and District: Administrative identifiers for the location of cultivation.
4. Crop: The specific crop grown (e.g., Arhar/Tur, Moong, Urad).
5. Season: Kharif or Rabi, indicating the cultivation period.
6. Area: The area sown under the crop in hectares.
7. Production: The total output in metric tonnes.
8. Yield: Calculated as production per unit area (kg/ha)

3. Data Cleaning and Standardization

- To ensure accuracy and usability:
- Inconsistent or missing values were either excluded or handled through imputation (e.g., if yield was missing but area and production were present, yield was calculated).
- Crop names and seasonal tags were standardized to avoid duplication or spelling inconsistencies.
- Duplicate records (if any) were removed after verifying the values.

4. Derived Metrics

- Yield (kg/ha) was derived using the formula:
- $\text{Yield} = (\text{Production} / \text{Area}) \times 1000$. This standardizes yield for comparative analysis.

5. Analytical Scope

- This structured dataset enables:
- Time-series analysis of crop productivity.
- District-wise and crop-wise performance comparisons.
- Anomaly detection (spikes or drops in yield).
- Agricultural forecasting and spatial trend mapping.

6. Limitations

- The dataset may not capture all crops uniformly across all years and districts.
- External factors such as weather anomalies, pest outbreaks, or local irrigation issues are not directly included but may influence yield variations.

Variable Explanation

- **State Name:** Name of the Indian state where the crop was cultivated.
- **District Name:** Name of the district within the state.
- **Crop Year:** The year in which the crop was harvested.
- **Season:** The agricultural season (e.g., Kharif, Rabi) in which the crop was grown.
- **Crop:** The type of crop cultivated (e.g., Rice, Wheat).
- **Area:** Land area (in hectares) used for growing the crop.
- **Production:** Total crop production (in tonnes) for the given area.

Data Cleaning and Preprocessing

- We have used the group-wise median method to fill the missing values in the variable 'Production'.
- This method utilizes the median value of 'Production' grouped by each unique combination of 'Crop' and 'Season'.
- For each group (i.e., each Crop-Season pair), the missing values in 'Production' are replaced with the median of the non-missing values within that group.
- This helps in maintaining the central tendency of the data specific to crop and seasonal characteristics, thereby ensuring more context-relevant imputation.

Statistical Analysis

Assessing the consistency of crop yields and productions over time

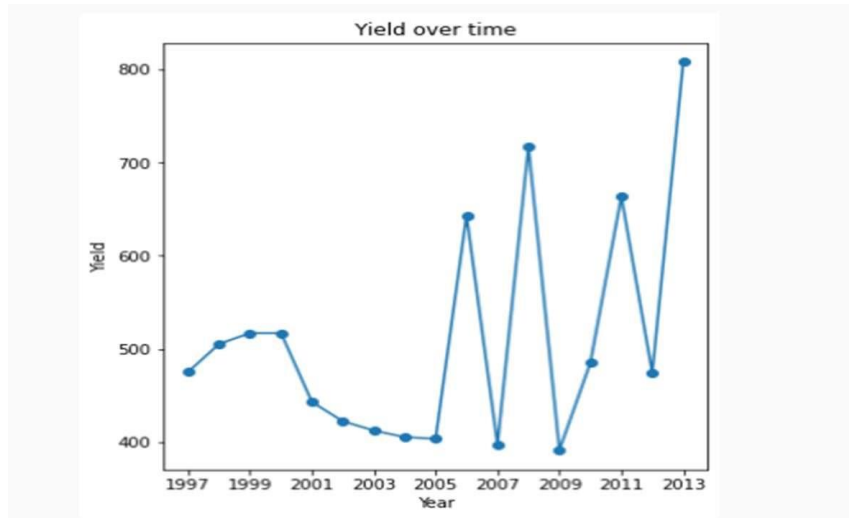


Figure 1: Trend in Crop Yield from 1997 to 2013

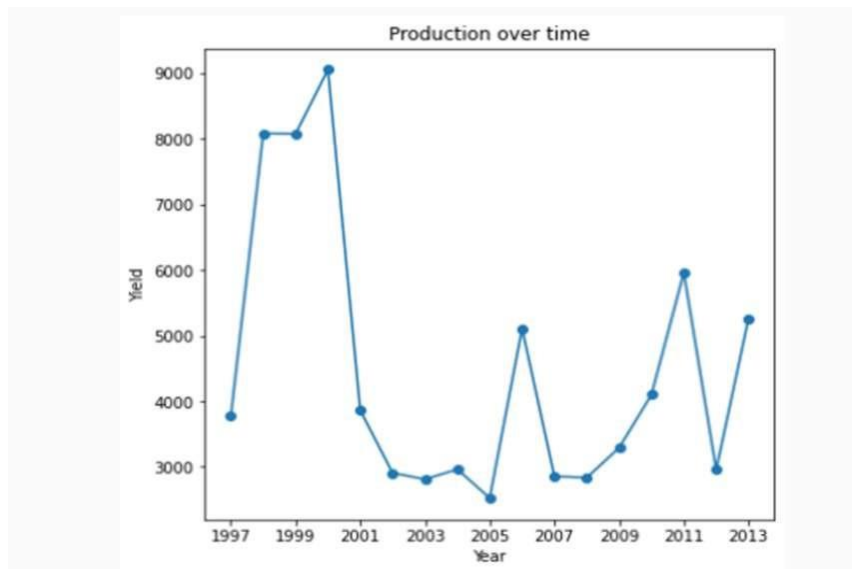


Figure 2: Trend in Crop Production from 1997 to 2013

The two figures illustrate the trends in crop yield and total production from 1997 to 2013. In Figure 1, crop yield shows notable fluctuations, with a decline until the mid-2000s and a sharp increase after 2009, indicating improved productivity. Figure 2 displays inconsistent production levels, with a peak around 1999, a significant drop in the early 2000s, and moderate recovery in later years.

Shape of the distribution of yield for each crop

Crop:Arhar/Tur
mean:712.94,median:678.00,skewness:4.51
shape of distribution:symmetrical
number of outliers:15
Interpretation:Yields are consistent .Indicates stable condition and practices

Crop:Ash Gourd
mean:0.00,median:0.00,skewness:nan
shape of distribution:left-skewed
number of outliers:0
Interpretation:Most yields are high with a few lower values. Could point to isolated problem like pests, drought or poor plots

Crop:Beans & Mutter(Vegetable)
mean:0.00,median:0.00,skewness:nan
shape of distribution:left-skewed
number of outliers:0
Interpretation:Most yields are high with a few lower values. Could point to isolated problem like pests, drought or poor plots

Crop:Bhindi
mean:7231.55,median:7008.00,skewness:1.82
shape of distribution:symmetrical
number of outliers:10
Interpretation:Yields are consistent .Indicates stable condition and practices

Crop:Bottle Gourd
mean:0.00,median:0.00,skewness:nan
shape of distribution:left-skewed
number of outliers:0
Interpretation:Most yields are high with a few lower values. Could point to isolated problem like pests, drought or poor plots

Crop:Cabbage
mean:60286.95,median:63529.00,skewness:-1.03
shape of distribution:symmetrical
number of outliers:2
Interpretation:Yields are consistent .Indicates stable condition and practices

Crop:Drum Stick
mean:0.27,median:0.00,skewness:7.42
shape of distribution:Right-skewed
number of outliers:1
Interpretation:Most yields are lower with a few higher values. Could indicate resource inequality or isolated success

Crop:Garlic
mean:5373.49,median:5626.00,skewness:-1.79
shape of distribution:symmetrical
number of outliers:10
Interpretation:Yields are consistent .Indicates stable condition and practices

Crop:Gram
mean:1348.62,median:591.50,skewness:6.31
shape of distribution:Right-skewed
number of outliers:13
Interpretation:Most yields are lower with a few higher values. Could indicate resource inequality or isolated success

Crop:Horse-gram
mean:4794.58,median:452.50,skewness:6.35
shape of distribution:Right-skewed
number of outliers:24
Interpretation:Most yields are lower with a few higher values. Could indicate resource inequality or isolated success

Crop:Lab-Lab

mean:0.00,median:0.00,skewness:nan

shape of distribution:left-skewed

number of outliers:0

Interpretation:Most yields are high with a few lower values. Could point to isolated problem like pests, drought or poor plots

Crop:Moong(Green Gram)

mean:1154.39,median:455.00,skewness:22.27

shape of distribution:Right-skewed

number of outliers:12

Interpretation:Most yields are lower with a few higher values. Could indicate resource inequality or isolated success

Crop:Other Kharif pulses

mean:2718.67,median:202.00,skewness:11.58

shape of distribution:Right-skewed

number of outliers:16

Interpretation:Most yields are lower with a few higher values. Could indicate resource inequality or isolated success

Crop:Other Vegetables

mean:0.40,median:0.00,skewness:7.55

shape of distribution:Right-skewed

number of outliers:1

Interpretation:Most yields are lower with a few higher values. Could indicate resource inequality or isolated success

Crop:Pulses total

mean:442.04,median:445.00,skewness:0.28

shape of distribution:symmetrical

number of outliers:1

Interpretation:Yields are consistent .Indicates stable condition and practices

Crop:Ribed Guard

mean:0.00,median:0.00,skewness:nan

shape of distribution:left-skewed

number of outliers:0

Interpretation:Most yields are high with a few lower values. Could point to isolated problem like pests, drought or poor plots

Crop:Urad

mean:529.85,median:480.00,skewness:16.66

shape of distribution:symmetrical

number of outliers:7

Interpretation:Yields are consistent .Indicates stable condition and practices

Crop-wise Statistical Comparisons

F-statistic: 40.07048941234184
p-value: 1.3778792078440907e-112

The F-statistic value of **40.07** indicates that there is a substantial difference in the mean yields among different crops when compared to the variation within each crop group. The extremely small p-value of **1.37788×10^{-112}** confirms that this difference is **statistically significant**, meaning it is highly unlikely to have occurred by chance. Since the p-value is far below the standard threshold of 0.05, we can confidently reject the null hypothesis and conclude that **there are significant differences in average yields between at least some of the crops** in the dataset.

Highest and Lowest Yielding Crops

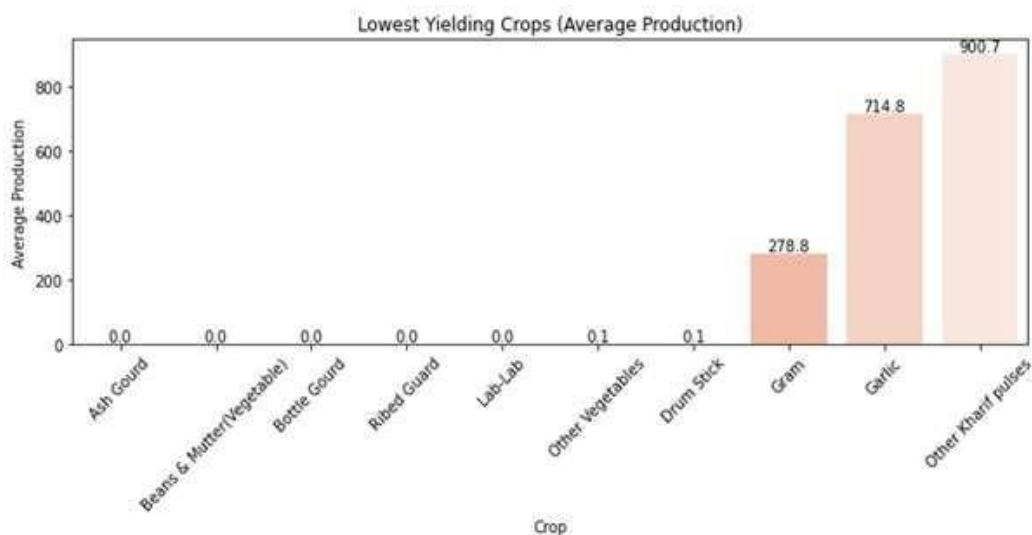


Figure 3: Lowest Yielding Crops based on the Average Production

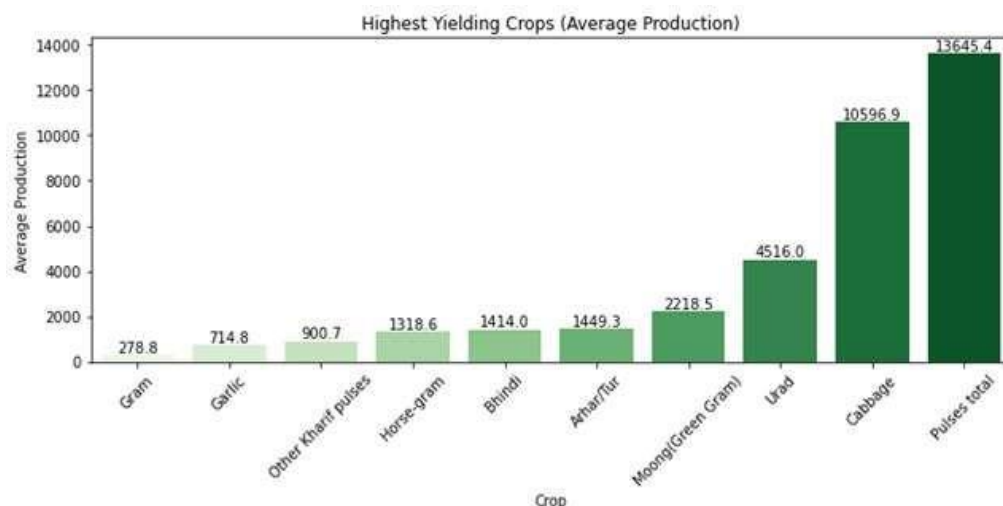


Figure 4: Highest Yielding Crops based on the Average Production

These charts compare crops based on their average production. Figure 3 shows that crops like ash gourd, bottle gourd, and ridge gourd have very low yields, with garlic and floral plants performing slightly better among the lowest. Figure 4 highlights crops such as cauliflower and paddy as the highest yielders, with paddy showing the maximum average production of 13,445.4 units, indicating its dominance in overall agricultural output.

Trend and Growth Analysis Over Time

The top 10 crops with Highest Increase and Lowest Decrease in Production

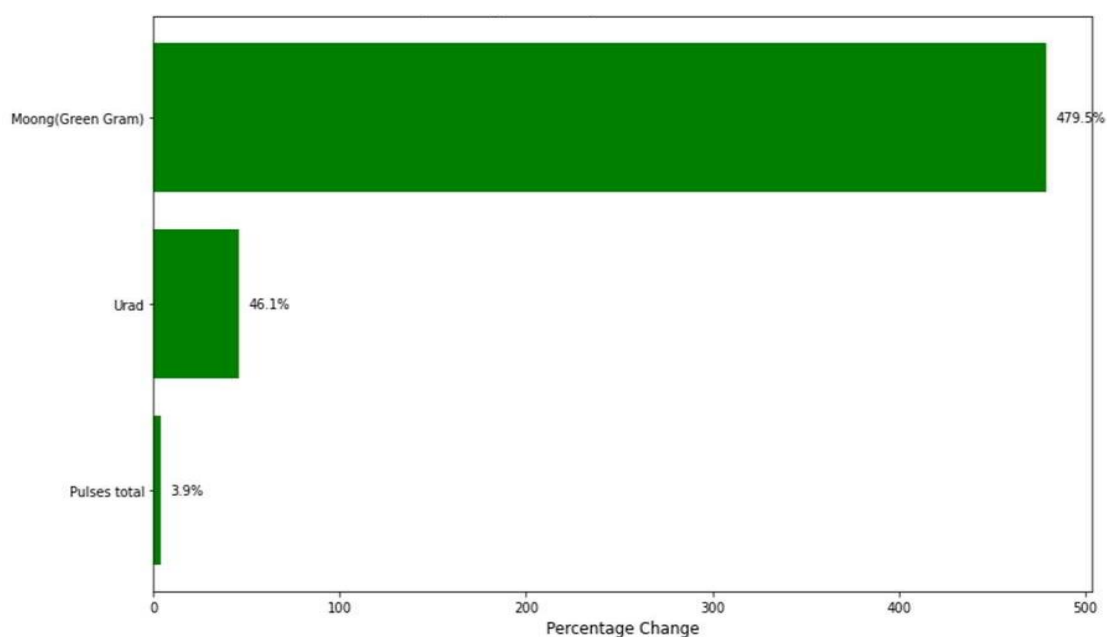


Figure 5: Top 3 Crops with Highest increase in Production

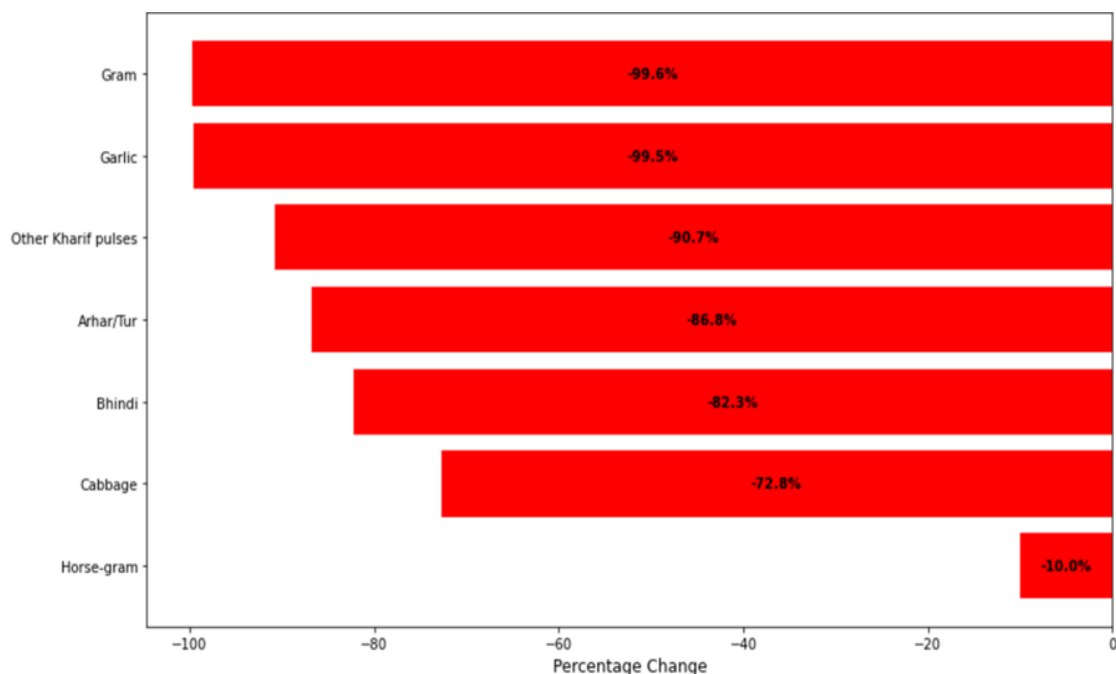


Figure 6: Top 7 Crops with Highest decrease in Production

Figures 5 and 6 show the percentage change in production of selected crops over time. Figure 5 highlights crops with the highest increase in production, where Moong (Green Gram) recorded a substantial rise of 479.3%, followed by Urad at 46.1%, and Pulses total with a modest increase of 3.9%. In contrast, Figure 6 displays crops with significant decreases in production, with Gram and Garlic showing nearly 50% decline, followed by other crops like Other Kharif pulses, Arhar/Tur, Brinjal, Cabbage, and Horse Gram, indicating downward trends in their cultivation or yield.

Crop-specific Performance Analysis

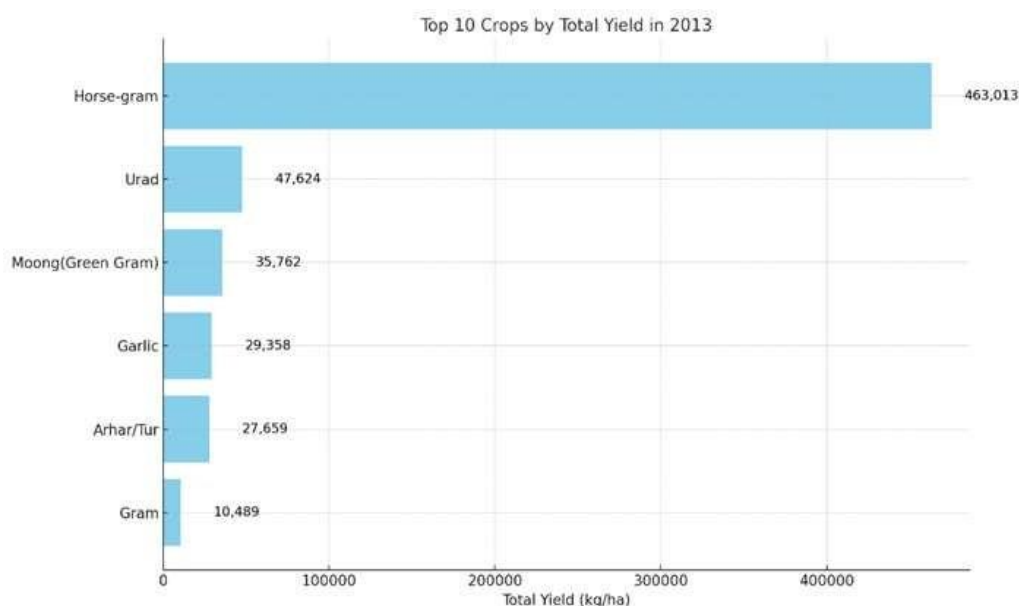


Figure 7: Top 10 Crops by Total Yield in 2013

Figure 7 presents the total yield (in kg/ha) of the top crops in the year **2013**. Among them, **Horse Gram** stands out significantly with a yield of **463,013 kg/ha**, far surpassing other crops. It is followed by **Urad** (47,624 kg/ha), **Moong (Green Gram)** (35,762 kg/ha), **Garlic**, **Arhar/Tur**, and **Gram**. This analysis indicates that **Horse Gram was the highest-yielding crop in 2013**, contributing a major portion of the total crop yield for that year.

Greatest percentage increase in production over the time period

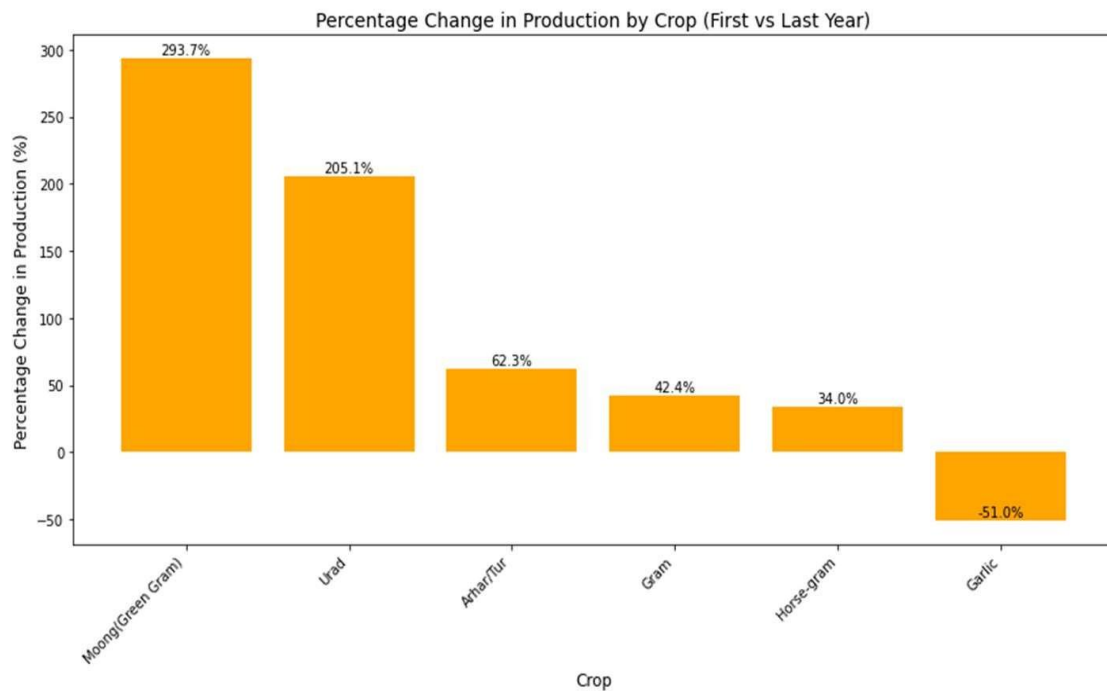


Figure 8: Percentage change in Crop Production over the time period

Figure 8 displays the percentage change in production for selected crops between the first and last recorded years.

Among the crops analyzed, **Moong (Green Gram)** shows the **greatest increase in production** with a **293.66%** rise, followed by **Urad** at **205.11%** and **Arhar/Tur** at **62.35%**. Other crops like **Gram** and **Horse Gram** also show moderate increases.

On the other hand, **Garlic** experienced a **decline of 50.95%** in production over the same period.

This analysis highlights the crops with the most significant growth and decline, indicating shifts in agricultural patterns or productivity over time.

Lowest yield in both the initial and latest years

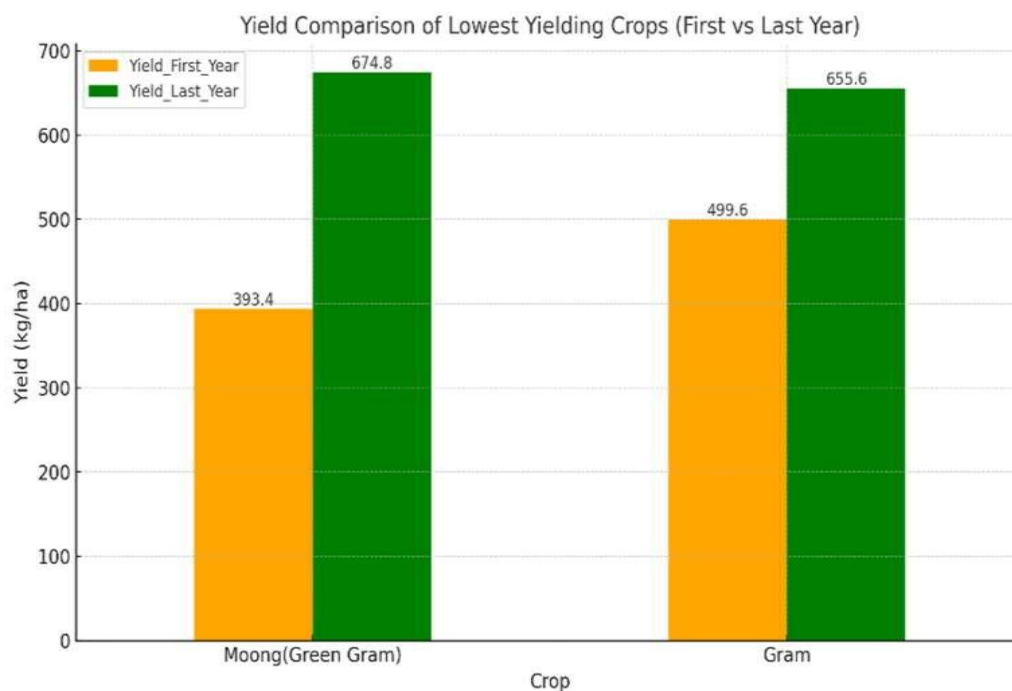


Figure 9: Yield Comparison of Lowest Yielding Crops between First and Last Year

Figure 9 illustrates the changes in yield for two crops that were initially identified as among the lowest yielders: Moong (Green Gram) and Gram. The chart compares the yield values (in kg/ha) recorded during the first year of observation with those in the final year of the study period. For Moong (Green Gram), the yield improved significantly from 393.4 kg/ha in the initial year to 674.8 kg/ha in the final year. This represents an approximate 71.5% increase in yield over time, indicating substantial progress in the crop's productivity, possibly due to better farming practices, improved seed quality, or favourable climatic conditions. In the case of Gram, the yield increased from 499.6 kg/ha to 655.6 kg/ha, marking a 31.2% increase. Although the growth in Gram's yield is comparatively lower than that of Moong, it still reflects a positive trend and improved efficiency in cultivation.

Efficiency & Productivity Metrics

	Crop	...	total_productivity_improvement_ratio
0	Arhar/Tur	...	-0.296058
1	Ash Gourd	...	NaN
2	Beans & Mutter(Vegetable)	...	NaN
3	Bhindi	...	1.216832
4	Bottle Gourd	...	NaN
5	Cabbage	...	0.999806
6	Drum Stick	...	NaN
7	Garlic	...	-inf
8	Gram	...	1.000404
9	Horse-gram	...	0.999082
10	Lab-Lab	...	NaN
11	Moong(Green Gram)	...	-4.862392
12	Other Kharif pulses	...	1.005462
13	Other Vegetables	...	NaN
14	Pulses total	...	-0.169364
15	Ribed Guard	...	NaN
16	Urad	...	-0.057385

Figure 10: Efficiency and Productivity Metrics of Crops

Figure 10 presents the total productivity improvement ratio for various crops over the study period. This ratio reflects how much each crop's productivity has improved or declined overall.

- The crop with the highest improvement is Bhindi, with a productivity improvement ratio of 1.2168, indicating strong positive progress in yield or output efficiency.
- On the other hand, Moong (Green Gram) shows the greatest decline, with a significantly negative ratio of -4.8624, highlighting a drop in productivity and the urgent need for agronomic or policy-level interventions.
- Crops such as Gram, Cabbage, and Horse Gram have ratios close to 1, suggesting stable or slightly improved productivity.
- Arhar/Tur, Pulses total, and Urad exhibit moderate to declining trends, indicating they may also require focused improvement strategies.

The analysis identifies pulses, especially Moong and Arhar/Tur, as key areas of concern, where productivity has either stagnated or declined, signaling a need for targeted support in terms of farming techniques, seed quality, irrigation, or market support.

Comparative Performance Between Crops

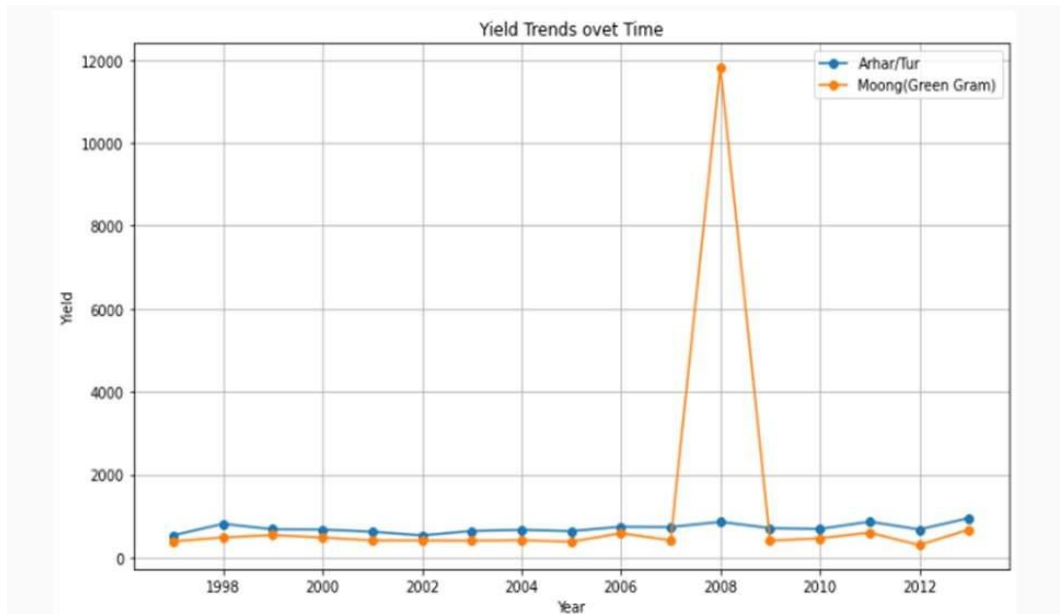


Figure 11: Comparative Yield Trends of Arhar/Tur and Moong over time

Figure 11 compares the yield trends of two crops — Arhar/Tur and Moong (Green Gram) across the years from the late 1990s to 2013. The line graph reveals that Moong experienced a sharp spike in yield around the year 2008, reaching an unusually high value. However, this appears to be an isolated peak rather than a sustained trend.

In contrast, Arhar/Tur shows a more stable and consistent upward trend in yield over the entire period. While its values are lower than Moong's peak, its performance is more reliable and gradual, suggesting a steadier improvement in productivity.

Thus, even though Moong had a dramatic rise in one year, Arhar/Tur demonstrates a more significant and meaningful long-term improvement in yield, making it the more consistently improving crop overall.

Compare the growth in yield and production between Gram and Urad

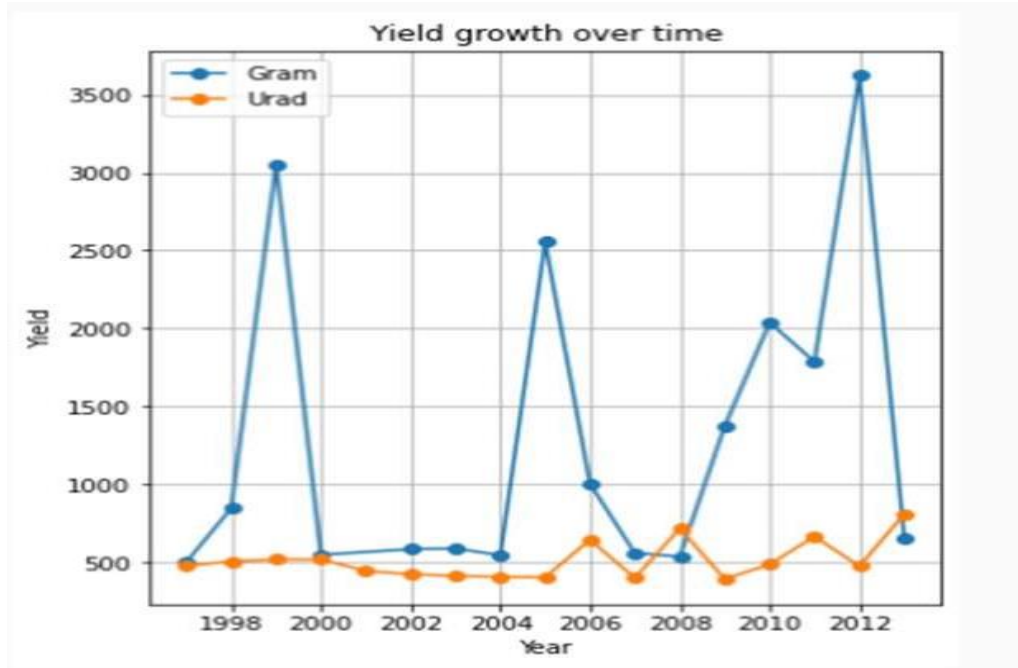


Figure 12: Yield Growth over time for Gram and Urad

Figure 12 illustrates the trend in yield growth between Gram and Urad from the initial to the final year of the study period. The data shows that:

Urad's yield increased significantly by 332.02 units, marked by sharp upward spikes in specific years (notably around 1999, 2005, and 2013).

Gram's yield saw a much lower increase of only 156.19 units, with relatively consistent but minimal changes across the years.

This indicates that Urad yield improved more significantly than Gram, suggesting that Urad may have benefitted from better agronomic practices, favourable climatic conditions, or targeted cultivation efforts.

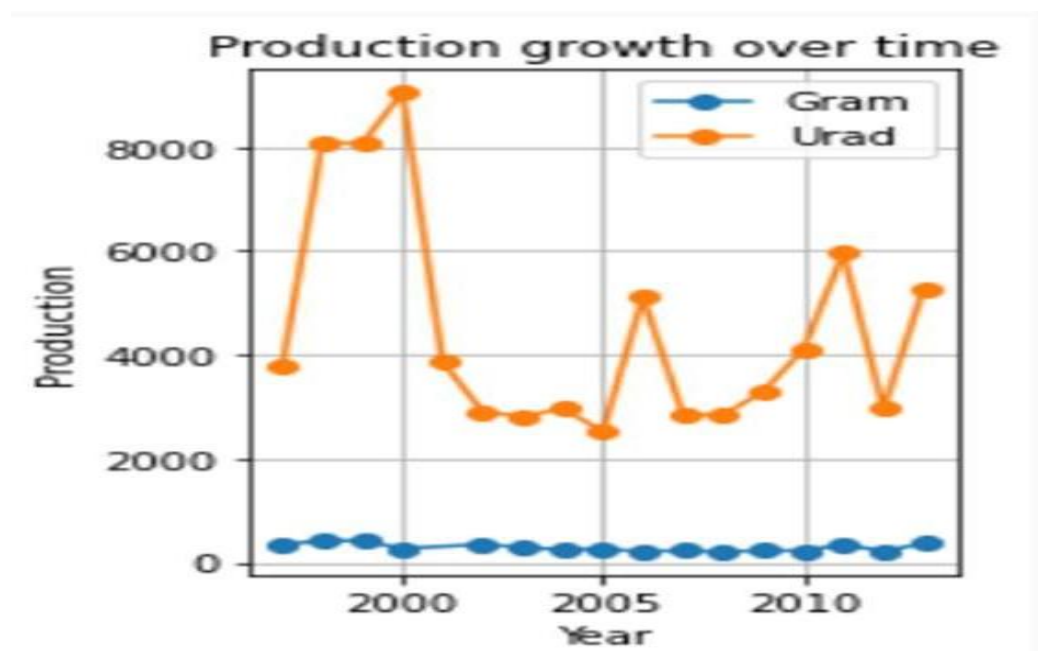


Figure 13: Production Growth over time for Gram and Urad

Figure 13 compares the production growth trends for Gram and Urad during the same period. The results clearly demonstrate:

Urad production grew dramatically by 1494.28 units, with high peaks and variability indicating responsiveness to favourable conditions or interventions.

In contrast, Gram production showed a negligible increase of just 23.10 units, remaining mostly flat over the years.

This analysis reveals that Urad experienced substantial growth in production, while Gram remained stagnant, reinforcing Urad's superior performance in terms of both output and yield potential.

Time Series Decomposition and Smoothing

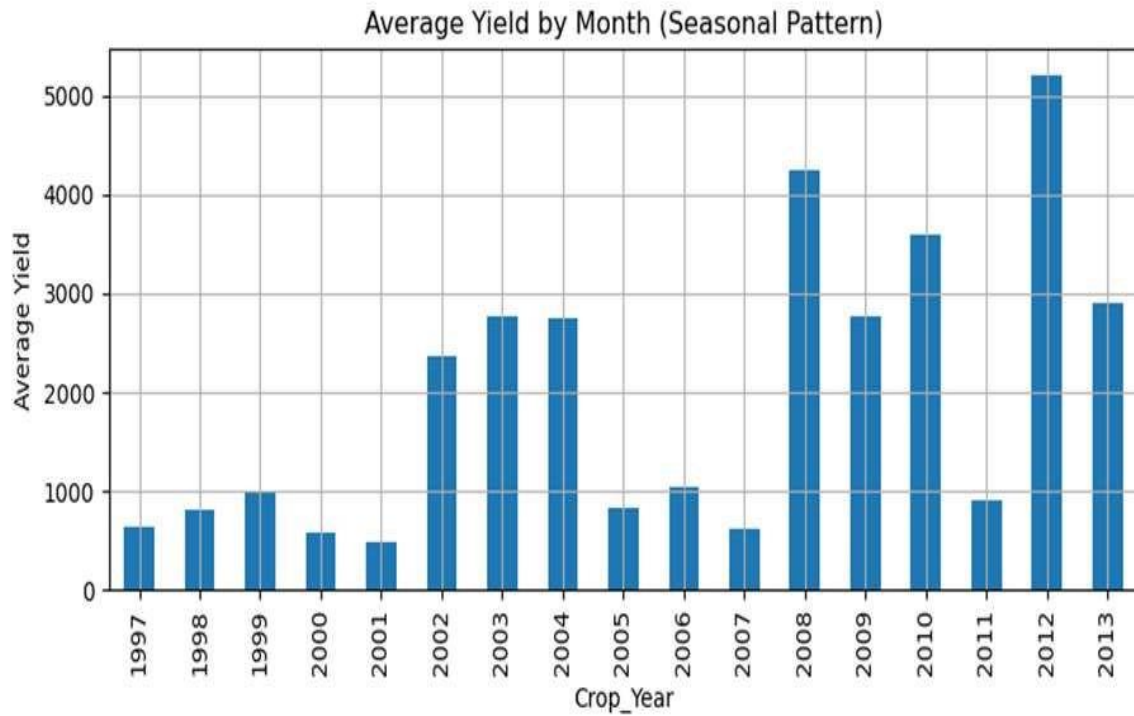


Figure 14: Average Yield by Crop Year (Seasonal Pattern)

The graph in Figure 14 illustrates the average crop yield from 1997 to 2013, highlighting clear seasonal patterns in agricultural productivity. Yield remained consistently low from 1997 to 2001, followed by a sharp increase between 2002 and 2004. Significant spikes are observed in 2008 and 2012, marking highly productive years. However, there is a noticeable decline in 2013. These fluctuations indicate that seasonal and environmental factors such as rainfall, climate, and input availability (e.g., fertilizers, labour, and machinery) have a substantial impact on yield outcomes. Understanding these trends is essential for optimizing sowing and harvesting schedules and for effective allocation of resources to enhance production and minimize risk.

Long-term productivity trends

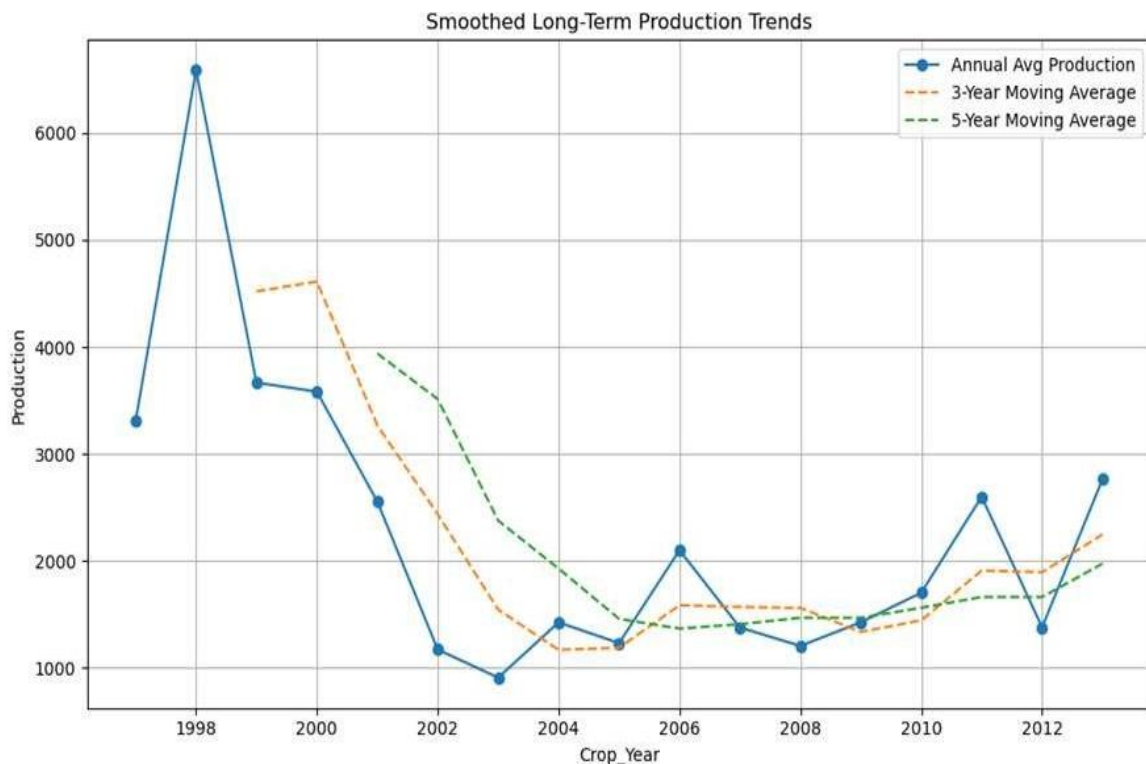


Figure 15: Smoothed Long Term Production Trends (1997-2013)

Figure 15 illustrates long-term productivity trends in agricultural production from 1997 to 2013, using annual average production data along with 3-year and 5-year moving averages to smooth out short-term fluctuations. The graph highlights a significant decline in productivity from 1998 to 2004, evident across both moving averages. This period of low productivity is followed by a gradual recovery after 2008, indicating a positive long-term trend.

The use of smoothing techniques helps to minimize the noise caused by year-to-year variations and provides a clearer picture of the underlying trends. This is valuable for agricultural planning, as it allows decision-makers to develop more stable long-term strategies for resource allocation, crop selection, and productivity improvement.

Abrupt Changes & External Factors

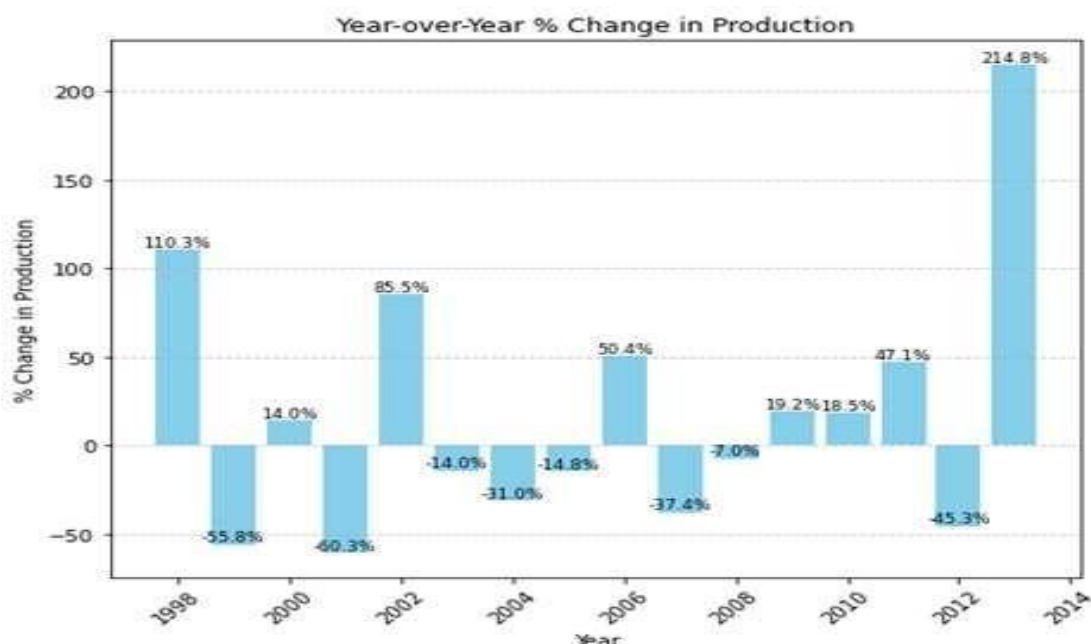


Figure 16a: Year-over-Year Percentage Change in Production (1998-2013)

Figure 16a presents the annual percentage changes in agricultural production between 1998 and 2013. It highlights how external conditions have driven abrupt year-to-year shifts.

Key highlights:

- The most dramatic increase occurred in 2013 (+214.8%), signaling an exceptional recovery or intervention.
- A major spike is also seen in 1998 (+110.3%), indicating a rebound after a poor year.
- Severe declines were recorded in 2000 (-55.8%), 2001 (-40.3%), and 2009 (-37.4%), likely due to unfavorable conditions or resource constraints.
- Other noticeable gains include 2005 (+50.4%), 2010 (+47.1%), and 2012 (+45.3%).

These fluctuations emphasize how sensitive production is to short-term factors like weather, pest pressure, and input access. Recognizing these abrupt changes can help in better planning, risk mitigation, and policy development.

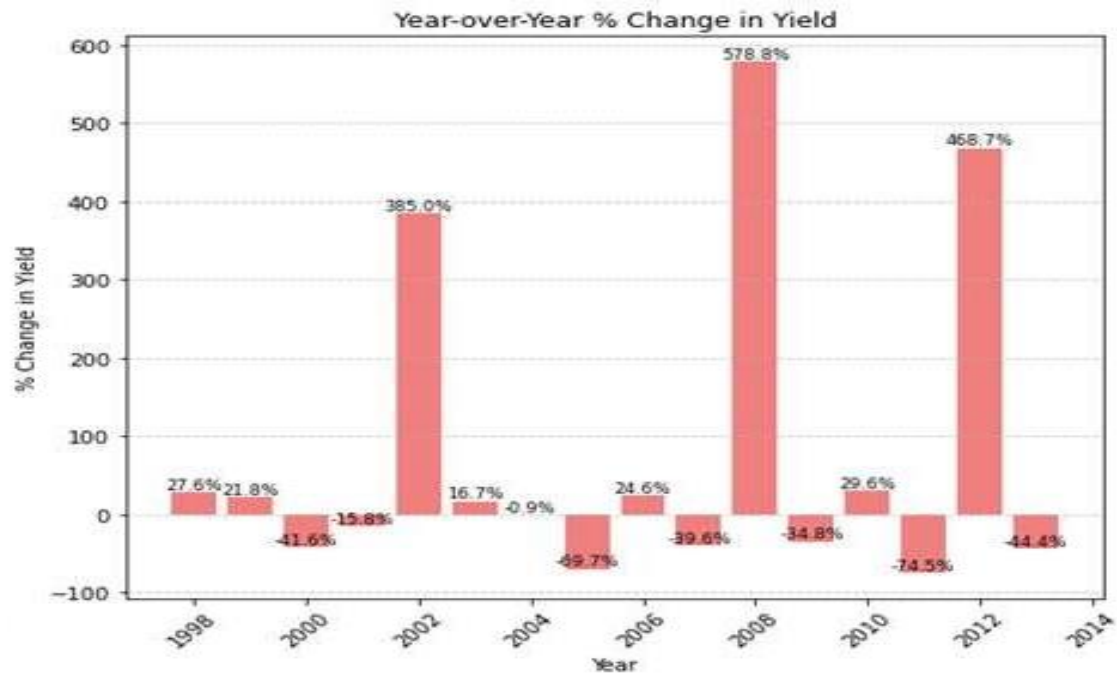


Figure 16b: Year-over-Year Percentage Change in Yield (1998-2013)

Figure 16b displays the annual percentage change in yield, further highlighting sudden spikes and dips due to environmental or management-related factors. Key observations include:

- Dramatic surges in yield during 2008 (578.8%) and 2012 (468.7%), suggesting exceptional conditions or interventions.
- Major declines in 2001 (-41.6%), 2005 (-49.2%), and especially 2013 (-71.5%) point to stress periods, possibly due to drought, poor inputs, or disease.

This chart underscores the sensitivity of yield to external influences and management practices.

Monitoring such changes supports better risk mitigation, contingency planning, and yield forecasting in agricultural systems

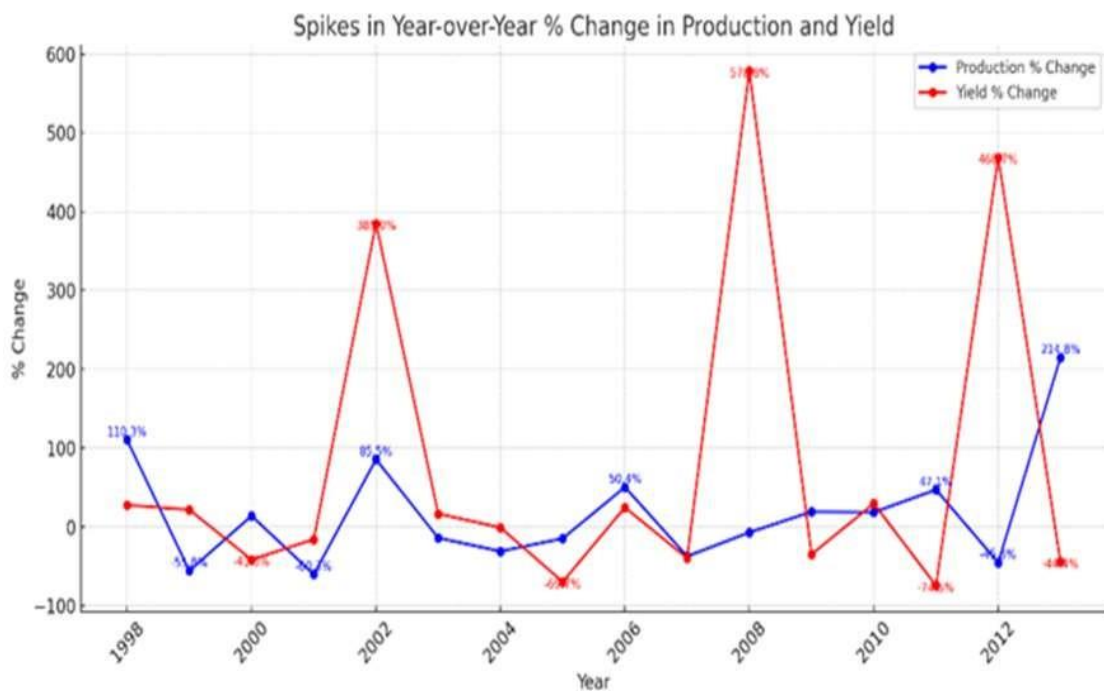


Figure 17: Spikes in Year-over-Year Percentage Change in Production and Yield (1998-2013)

Figure 17 illustrates spikes in year-over-year % changes for both production and yield, helping identify years with extreme agricultural fluctuations. The graph uses line plots to compare the magnitude and timing of these changes over time.

Key insights:

Spikes in Yield:

The years 2002, 2008, and 2012 exhibit extreme positive spikes in yield percentage change, with 2008 reaching the highest at approximately 578.8%.

Negative Production Changes:

Significant declines in production are seen in 1999, 2001, and 2012, highlighting periods of stress or disruption such as unfavorable weather, pest issues, or input shortages.

Production Surge in 2013:

Despite a drop in yield that year, 2013 saw a massive 214.8% increase in production, likely driven by a substantial expansion in cultivation area or external intervention.

By comparing yield and production trends together, the figure helps to differentiate between yield-driven changes (efficiency/productivity per unit area) and those caused by changes in total cultivated area.

Suggest potential external factors (e.g., weather events, pest outbreaks, policy changes) that could explain these abrupt changes

Huge Production Spike (+110%) Possible factors:

- Good monsoon: The Indian Meteorological Department (IMD) reported a near-normal monsoon in 1998.
- Government incentives: Possible increase in support prices or input subsidies.

1999 – Production Drop (–55%) Possible factors:

- Severe drought: 1999 was marked by a weak monsoon, leading to water stress.
- Crop loss due to pests in several regions, especially pulses and oilseeds.

2000–2001 – Drop in Yield & Production Possible factors:

- Back-to-back dry spells.
- Soil degradation or pest resurgence may have lowered crop efficiency.
- Potential reduction in cultivated area or irrigation issues.

2013 – Huge Production Spike (+214%) but Yield Drop Possible causes:

- Massive increase in cultivated area, maybe due to government incentives.
- Yield dropped possibly due to newly cultivated marginal lands being less fertile or poorly managed.

Summary of Possible Influences

Type of Factor	Description
Weather	Droughts, floods, irregular rainfall patterns
Biological	Pest outbreaks, crop diseases
Policy	MSP changes, export bans, subsidies, crop insurance programs
Technology	Introduction of high-yielding seeds, mechanization, better irrigation
Data Factors	Abrupt yield spikes often indicate area anomalies or reporting errors

Table 1: Summary of Possible Influences on Agricultural Production and Yield

Table 1 presents a categorized summary of key external and internal factors that may influence abrupt changes in agricultural production and yield. These influences help explain the volatility observed in previous figures and support better interpretation of trends.

This structured overview helps stakeholders, analysts, and planners pinpoint potential root causes behind production/yield fluctuations and make more informed decisions in agriculture management.

Policy Implications

- Support pulses with declining trends using agronomic research and subsidies.
- Prioritize extension services for low-efficiency crops.
- Use time-series findings for disaster preparedness and resource allocation.
- Encourage adoption of high-efficiency practices in low-yielding areas.

Conclusion

This internship provided a valuable opportunity to bridge academic knowledge with real-world applications, specifically in the context of agricultural data analysis. Through hands-on involvement with actual crop production and yield data, the internship enabled the practical application of core statistical techniques in a professional setting.

One of the primary takeaways was the enhancement of data handling skills, particularly in data cleaning and preprocessing, which are essential when working with large and sometimes inconsistent datasets. Additionally, the internship offered in-depth exposure to various statistical tests and tools, improving proficiency in hypothesis testing and inferential techniques.

A major focus was placed on growth analysis, allowing for the examination of temporal trends in crop production and yield. This included working with time series data, where interpreting seasonal patterns, year-on-year variations, and identifying external shocks (such as weather events or policy changes) became crucial components of the analysis. The experience helped solidify the understanding of how statistical tools can be used to inform agricultural decision-making and planning.

Moreover, the internship highlighted the practical challenges often faced in statistical analysis, such as issues of data quality, missing values, and external variability. Confronting these issues firsthand emphasized the importance of critical thinking, adaptability, and domain knowledge when interpreting results.

Overall, this internship not only strengthened technical skills but also deepened the appreciation of statistics as a vital instrument in addressing real-world problems, especially in the context of agricultural development and policy evaluation.

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